KLE Technological University
Creating Value
Leveraging Knowledge

DEPARTMENT OF BACHELOR COMPUTER APPLICATION

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Project Title : Restaurant Search and Ranking System

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1.1 Introduction

Requirement analysis is the process of understanding what a system should do and what users need from it. It helps us identify the main features and requirements needed to develop the software.

The Restaurant Searching and Ranking System is developed to help users find suitable restaurants easily. Nowadays, many restaurants are available in different locations, and each restaurant offers different types of food, prices, and ratings. Because of this, users may find it difficult to choose the right restaurant.

This system makes restaurant searching easier by allowing users to search for restaurants based on their location, food type, price, and ratings. It collects restaurant information from a database and displays the restaurants that match the user's requirements.

The system also ranks restaurants according to their ratings and selected preferences. This helps users compare different restaurants and choose a suitable one without spending too much time searching manually.

For example, if a user wants to find a South Indian restaurant within a budget of ₹200, the system can display restaurants that match these requirements. It can then arrange the results according to their ratings.

The main purpose of the system is to save users' time, make restaurant searching simple, and help them choose suitable restaurants.

2.2 Objectives of Requirement Analysis

The main objectives of requirement analysis are:

1. Understand user needs: Identify what users expect from the restaurant searching system.
2. Easy searching: Allow users to search for restaurants by name and location.
3. Apply filters: Help users find restaurants based on cuisine, price, and ratings.
4. Rank restaurants: Display matching restaurants in a suitable order based on their ratings and preferences.
5. Store information: Maintain restaurant details in a database.
6. Save time: Reduce the time users spend searching for restaurants.
7. Easy to use: Provide a simple interface that users can understand without difficulty.

These objectives help define the main purpose and expected features of the system.

2.3 Functional Requirements

Functional requirements describe the main features and functions that the system must provide to users.

2.3.1 Restaurant Search

The system allows users to search for restaurants by entering a restaurant name or location. It checks the available information in the database and displays matching restaurants.

For example, if a user searches for restaurants in Hubballi, the system displays restaurants available in that location.

2.3.2 Search by Cuisine

Different users prefer different types of food. The system allows users to search for restaurants based on cuisine, such as South Indian, North Indian, Chinese, and Italian.

For example, if a user selects Chinese cuisine, the system displays restaurants that offer Chinese food.

2.3.3 Filter by Price

Users have different budgets when choosing restaurants. The system allows users to find restaurants according to their preferred price range.

For example, a user can search for restaurants where the average meal cost is below ₹200. The system displays restaurants that match the selected budget.

2.3.4 Filter by Ratings

Restaurant ratings help users understand how other customers have evaluated a restaurant.

The system allows users to view ratings and, if implemented, filter restaurants according to a minimum rating.

For example, a user can select restaurants with ratings of 4 stars or above. The system displays restaurants that satisfy this condition.

2.3.5 Restaurant Ranking

Restaurant ranking is one of the main features of the system.

The system arranges matching restaurants according to their ratings and other selected preferences. Restaurants with higher ratings can appear before restaurants with lower ratings when rating is the main ranking factor.

For example, if three restaurants have ratings of 4.5, 4.2, and 3.8, the system displays them in descending order of rating.

This feature helps users compare restaurants and choose a suitable option.

2.3.6 Display Restaurant Details

The system displays important information about each restaurant.

The information may include:

- Restaurant name
- Location and address
- Type of cuisine
- Price range
- Customer rating
- Reviews, if available

This information helps users understand the available options before selecting a restaurant.

2.3.7 Database Management

The system stores restaurant information in a database. Whenever a user searches for a restaurant, the system retrieves the required information from the database.

The database can be updated when new restaurants are added or existing restaurant details change.

2.3.8 Display Search Results

After processing the user's search, the system displays the matching restaurants.

The results are arranged according to the implemented ranking method. If no restaurant matches the user's requirements, the system displays a message asking the user to try different search criteria.

This makes the searching process simple and understandable.

2.4 Non-Functional Requirements

Non-functional requirements describe how well the system should work.

2.4.1 Performance

The system should display search results quickly. Users should not have to wait for a long time to find restaurants.

2.4.2 Usability

The system should be simple and easy to use. Search boxes, filter options, and restaurant details should be clearly displayed.

2.4.3 Reliability

The system should work correctly and display results based on the information available in the database.

2.4.4 Security

The system should protect its database and prevent unauthorized access. User inputs should be checked before processing.

2.4.5 Maintainability

The system should be easy to update. Developers should be able to add new restaurants, change existing details, and improve the ranking method when required.

2.4.6 Compatibility

The system should work on commonly used web browsers, such as Google Chrome and Microsoft Edge.

2.4.7 Accuracy

The system should display restaurants that match the user's search conditions. The accuracy of the results depends on the information stored in the database.

2.5 Hardware Requirements

Hardware requirements describe the physical resources needed to develop and run the system.

The basic hardware requirements are given below.

Component	Requirement
Processor	Intel i3 or equivalent
RAM	4 GB or above
Storage	10 GB free space
Input Devices	Keyboard and Mouse
Output Device	Monitor

These resources are sufficient for developing and testing a basic restaurant searching system.

2.6 Software Requirements

Software requirements describe the technologies and tools used to develop the system.

Component	Software
Operating System	Windows 10/11 or Linux
Frontend	HTML, CSS, JavaScript
Backend	Python with Flask
Database	MySQL
Web Browser	Google Chrome or Microsoft Edge
Code Editor	Visual Studio Code

Frontend: HTML, CSS, and JavaScript are used to create the website and provide an interface for users.

Backend: Python with Flask processes search requests, applies filters, and manages the ranking process.

Database: MySQL stores restaurant information and provides the required data when users search for restaurants.

Code Editor: Visual Studio Code is used to write and manage the project code.

2.7 Database Requirements

The database stores the information required for searching and ranking restaurants.

The main details stored in the database are:

Field Name	Description
Restaurant_ID	Unique ID of the restaurant
Restaurant_Name	Name of the restaurant
Location	Restaurant address or area
Cuisine	Type of food available
Price_Range	Average meal cost
Rating	Average customer rating
Reviews	Customer feedback, if available

For example, when a user searches for restaurants in a particular location, the system checks the location field and displays matching restaurants.

Similarly, the cuisine and price fields help the system filter restaurants according to the user's preferences. The rating field helps the system rank restaurants.

The database plays an important role because the system depends on stored information to generate search results.

2.8 System Modules

The system is divided into different modules. Each module performs a specific task.

2.8.1 User Interface Module

This module provides the interface through which users interact with the system. It contains search boxes, filter options, buttons, and restaurant details.

2.8.2 Search Module

This module searches for restaurants based on the information entered by users, such as restaurant name or location.

2.8.3 Filtering Module

This module filters restaurants according to the selected cuisine, price range, and ratings.

2.8.4 Ranking Module

This module arranges the matching restaurants according to their ratings and selected preferences. It helps users identify suitable restaurants easily.

2.8.5 Database Module

This module stores and retrieves restaurant information. It provides the data required by the search and ranking modules.

2.8.6 Results Module

This module displays the final restaurant list to users. It presents the restaurant details in an organized format so users can compare the available options.

All these modules work together to provide a simple restaurant searching and ranking service.

2.9 Assumptions and Limitations

Assumptions

The system is developed based on the following assumptions:

1. Restaurant information is available in the database.
2. Users enter valid search details.
3. Restaurant ratings and prices are stored correctly.
4. Users access the system through a web browser.
5. The system has access to the database while processing searches.

Limitations

The system may have the following limitations:

1. Search results depend on the information stored in the database.
2. Incorrect restaurant information may affect the results.
3. Restaurants that are not stored in the database cannot be displayed.
4. Real-time restaurant availability may not be provided.
5. Online booking and payment facilities are not included unless they are implemented separately.

These limitations describe the features and services available in the proposed system.

2.10 Feasibility Study

A feasibility study helps determine whether the system can be developed using the available resources.

2.10.1 Technical Feasibility

The system can be developed using HTML, CSS, JavaScript, Python, Flask, and MySQL. These technologies provide the features required to build the website, store restaurant details, and implement search and ranking functions.

2.10.2 Economic Feasibility

The system can be developed at a low cost because the suggested technologies are available for basic development without requiring paid software licences.

A computer and the required development tools are sufficient to start the project.

2.10.3 Operational Feasibility

The system is easy to use because users can search for restaurants, apply filters, and view ranked results through a simple interface.

Users do not need advanced technical knowledge to operate the system.

2.10.4 Schedule Feasibility

The project can be developed step by step. The first step is to create the database and user interface. Next, the search and filtering functions can be developed, followed by the ranking module and testing.

This approach helps complete the project within the available academic schedule.

2.11 Stakeholders of the System

Stakeholders are the people who use or manage the Restaurant Searching and Ranking System. The main stakeholders are:

1. Customers

Customers use the system to search for restaurants based on location, food type, price, and ratings. It helps them find a suitable restaurant easily.

2. Restaurant Owners

Restaurant owners provide details about their restaurants, such as name, location, food type, and price. This helps customers get information about different restaurants.

3. Administrator

The administrator manages restaurant information. They can add, update, and delete restaurant details to keep the information correct.

4. Developer

The developer creates and maintains the system. They make sure that the search and ranking features work properly.

Cooperation Between Stakeholders

All stakeholders work together to make the system useful. Restaurant owners provide restaurant details, and the administrator manages this information. Customers use the system to search for restaurants and view their details. The developer maintains the system and fixes problems when needed.

Conclusion

Each stakeholder has an important role in the system. Their cooperation helps users find restaurants easily and view useful information.

2.11 Summary

The requirement analysis explains the main requirements of the Restaurant Searching and Ranking System.

The system allows users to search for restaurants, filter results based on cuisine, price, and ratings, view restaurant details, and compare ranked results.

The functional requirements describe what the system should do, while the non-functional requirements describe how well it should work. The hardware and software requirements identify the resources needed for development.

The database stores restaurant information, and the different system modules work together to process search requests and display suitable results.

Overall, the Restaurant Searching and Ranking System aims to make restaurant searching easier, save users' time, and help them select restaurants according to their needs.